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CSC 415 Operating Systems

File System Project

Github Repository Link:

Milestone 1:

Overview

For this milestone, the goal was to have our volume fully formatted and ready to be used. This means that instead of just designing the file system like we did before, we now actually created the structures on disk and made sure everything is placed correctly. Our program creates a disk file called volume.dat and initializes all the required components, including the Volume Control Block (VCB), the free space management system, and the root directory. At this point, the volume is ready for future operations like file creation and allocation.

Volume Control Block (VCB)

The VCB is written to block 0 of the disk and acts as the main reference for the entire file system.

It stores important information such as:

- block size
- total number of blocks
- number of free blocks
- location of the free space bitmap
- location of the root directory

We also included a magic number to help identify that the disk contains our file system. One thing we had to be careful about was making sure that all the values in the VCB match what is actually written on disk. For example, the free block count is calculated after reserving space for the VCB, bitmap, and root directory, so it reflects the correct number of free blocks.

Free Space Structure

We used a bitmap to manage free space. Each block on the disk is represented by one entry in the bitmap.

- 1 means the block is used
- 0 means the block is free

During initialization, we mark all blocks that are already in use, including:

- block 0 (VCB)
- block 1 (root directory)
- bitmap blocks

Everything else is marked as free. We also implemented the allocation idea, where the system can scan the bitmap and find a free block when needed. The release function was not required for this milestone, so we did not fully implement it yet. This approach is simple and efficient enough for this stage, and it matches what we described in our design.

Directory System

The root directory is stored at block 1 and is initialized with two entries:

- "." (current directory)
- ".." (parent directory)

Since this is the root directory, both entries point to the same block.

Each directory entry includes:

- name
- location
- permissions
- timestamps

We used time_t for timestamps instead of strings, which made things easier and more consistent compared to our earlier design. Right now the directory is very basic, but it sets up the structure needed for adding files later.

Hex Dump

We used the provided HexDump utility to verify the contents of the volume file.

From the dump, we can clearly see:

- the VCB in block 0
- the bitmap showing used and free blocks
- the root directory entries in block 1

This helped confirm that everything was written correctly and in the right order.

Team Contributions

Team Member	Work Done
Ahmed Rashad	Worked on write up, disk initialization, VCB setup, and debugging block writes
Junxi Lin	Worked on write up, bitmap logic, root directory setup, and testing

We mainly communicated through Discord and sometimes met in person. We met a few times during the week, especially when we were trying to debug issues. Instead of fully splitting everything, we worked on different parts but still helped each other when something wasnt working. This made it easier to fix mistakes faster.

Challenges and Issues

One of the main issues was calculating the correct size of the bitmap and making sure it aligned with block boundaries. Small mistakes here caused incorrect free block counts at first. Another issue was making sure each structure was written to the correct block. If something is off by even one block, it becomes really hard to debug. We also had to deal with struct alignment issues, which we fixed using #pragma pack (1) to avoid padding problems. Overall, most problems were small but had a big impact, so debugging took some time.

Screenshots:

```
student@student:~/File-System-Milestone-One$ make clean
rm -f fs_format volume.dat
student@student:~/File-System-Milestone-One$ make
gcc fs_format.c -o fs_format
student@student:~/File-System-Milestone-One$ make run
./fs_format
File system start work.
student@student:~/File-System-Milestone-One$ ./Hexdump/hexdumpAS volume.dat --start 0 --count 1
Dumping file volume.dat, starting at block 0 for 1 block:

000000: 39 30 00 00 00 02 00 00  E8 03 00 00 E4 03 00 00 | 90******?***?***
000010: 02 00 00 00 02 00 00 00  01 00 00 00 01 00 00 00 | ****************
000020: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000030: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000040: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000050: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000060: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000070: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000080: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000090: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0000A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0000B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0000C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0000D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0000E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0000F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000100: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000110: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000120: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000130: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000140: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000150: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000160: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000170: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000180: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000190: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0001A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0001B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0001C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0001D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0001E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0001F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
student@student:~/File-System-Milestone-One$ ./Hexdump/hexdumpAS volume.dat --start 2 --count 2
Dumping file volume.dat, starting at block 2 for 2 blocks:

000400: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000410: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 01 00 | ****************
000420: 00 00 01 00 00 00 57 2E  D7 69 00 00 00 00 57 2E | ******W.?i****W.
000430: D7 69 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ?i**************
000440: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000450: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000460: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000470: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000480: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000490: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0004F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000500: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000510: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000520: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000530: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000540: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000550: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000560: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000570: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000580: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000590: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0005F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
000600: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000610: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000620: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000630: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000640: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000650: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000660: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000670: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000680: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000690: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0006F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000700: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000710: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000720: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000730: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000740: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000750: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000760: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000770: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000780: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000790: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0007E0: 00 00 00 00 00 00 00 00  21 F7 01 00 00 00 00 00 | ********!?******
0007F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
```

```
student@student:~/File-System-Milestone-One$ ./Hexdump/hexdumpAS volume.dat --start 1 --count 1
Dumping file volume.dat, starting at block 1 for 1 block:

000200: 2E 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | .***************
000210: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000220: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000230: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000240: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000250: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000260: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000270: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000280: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000290: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0002F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

000300: 00 00 00 01 00 00 00 01  00 00 00 57 2E D7 69 00 | ***********W.?i*
000310: 00 00 00 57 2E D7 69 00  00 00 00 2E 2E 00 00 00 | ***W.?i****..***
000320: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000330: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000340: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000350: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000360: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000370: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000380: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
000390: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003A0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003B0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003C0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003D0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003E0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************
0003F0: 00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00 | ****************

student@student:~/File-System-Milestone-One$
```